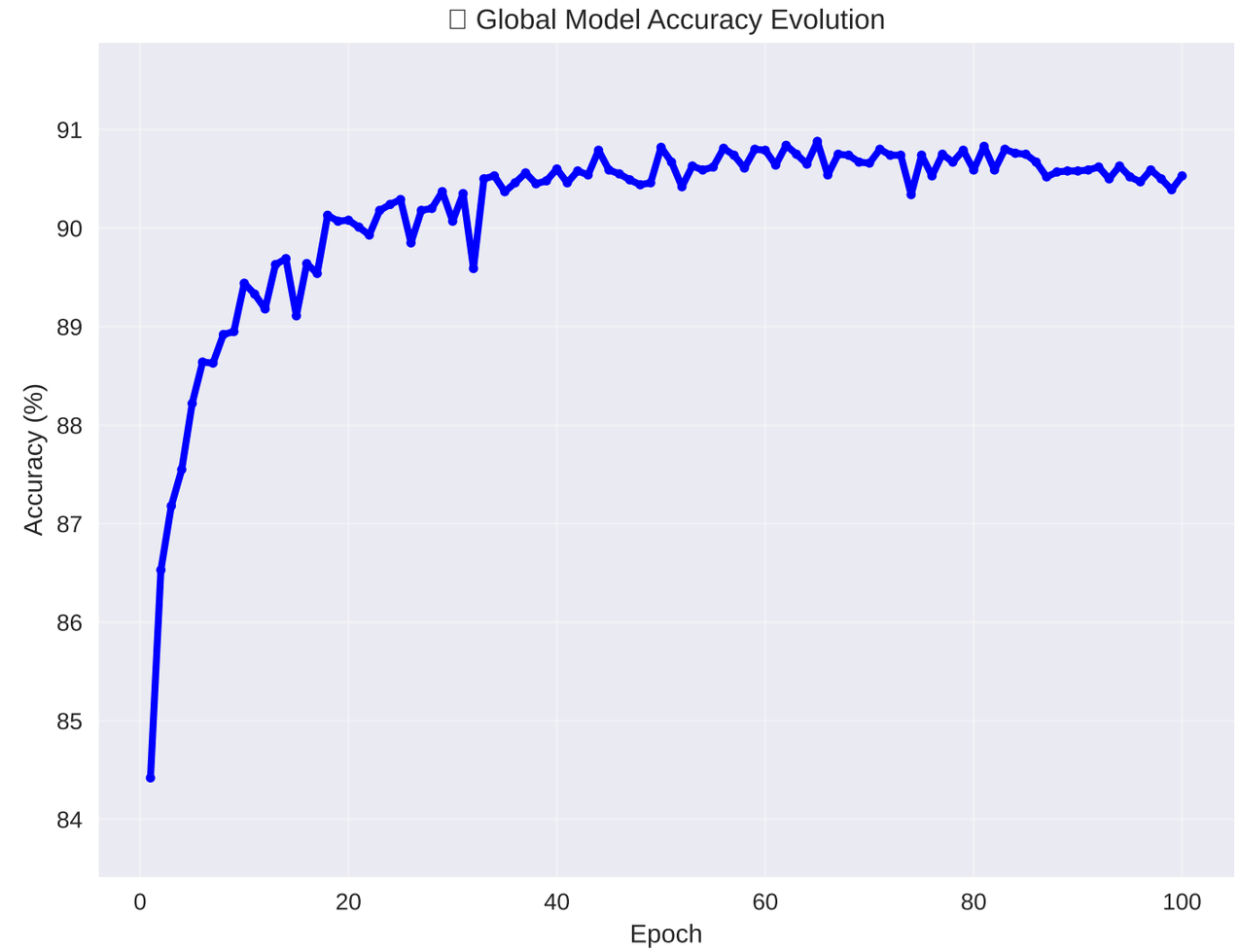
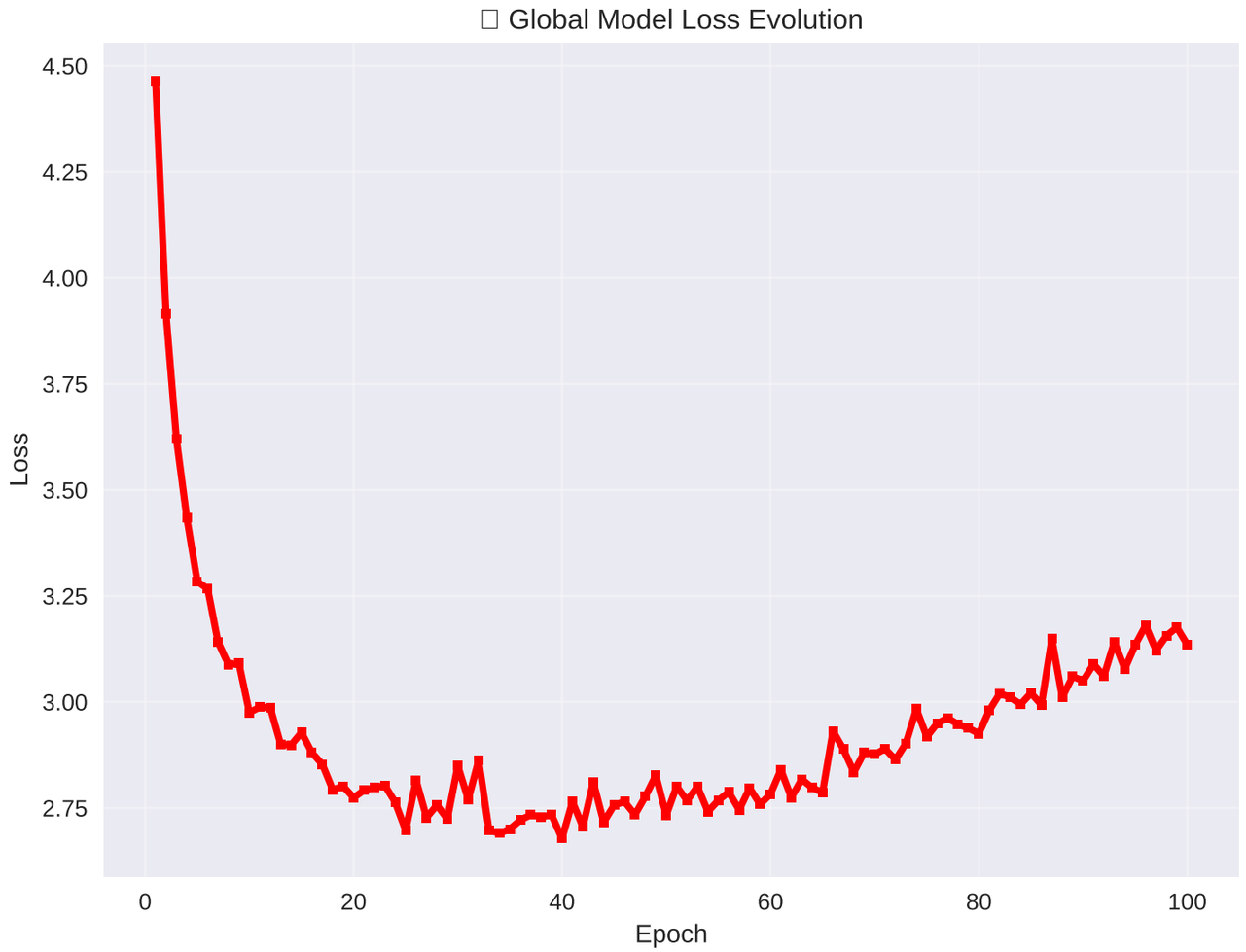
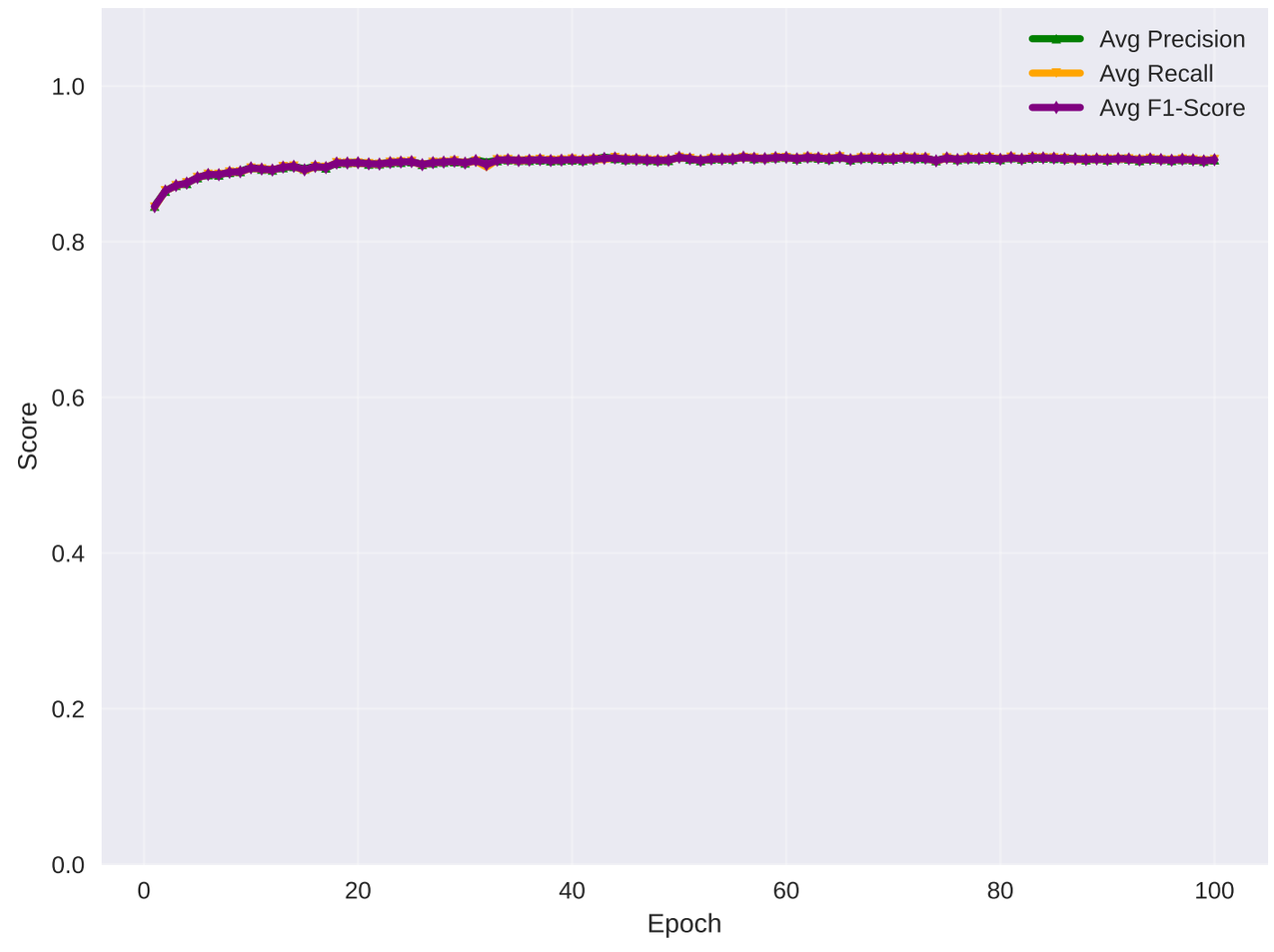
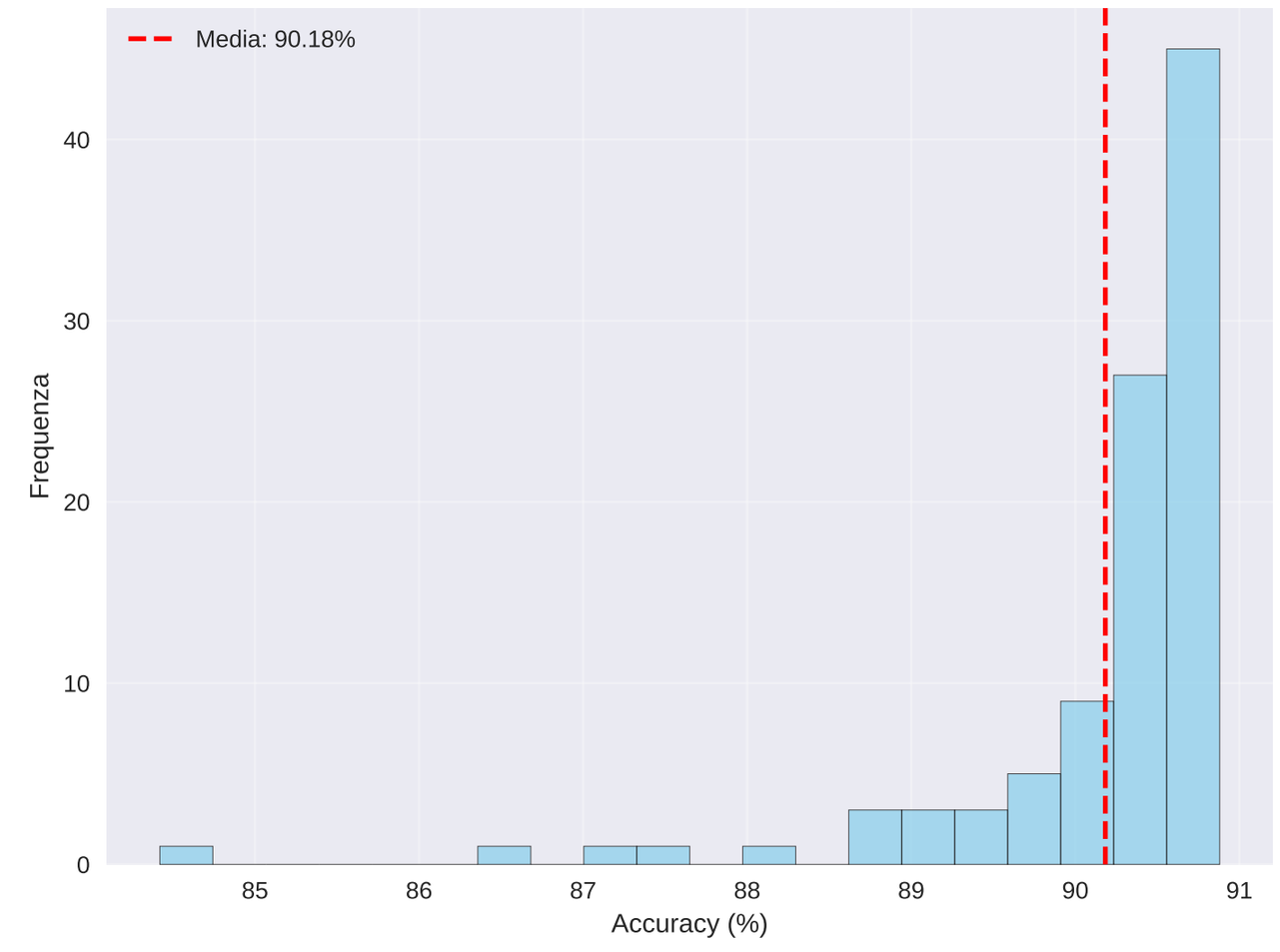
Federated Learning - Main Metrics Overview



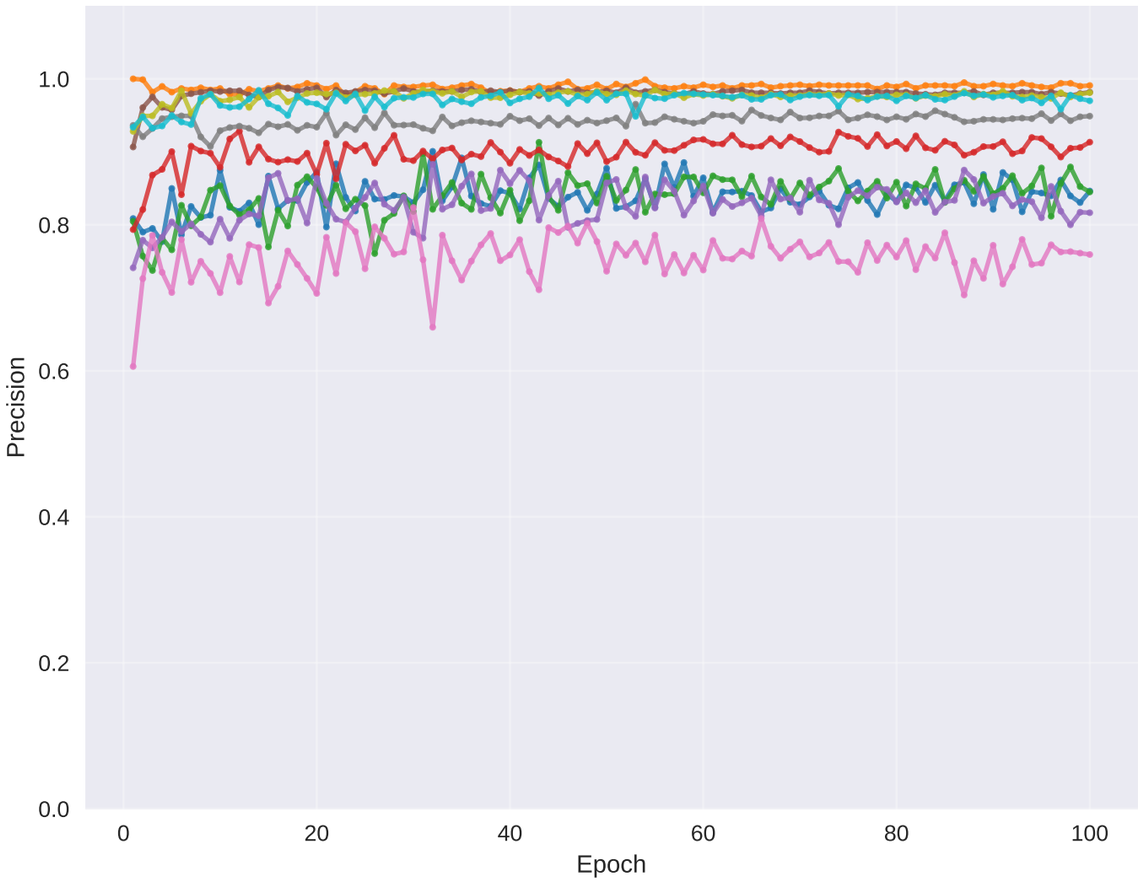




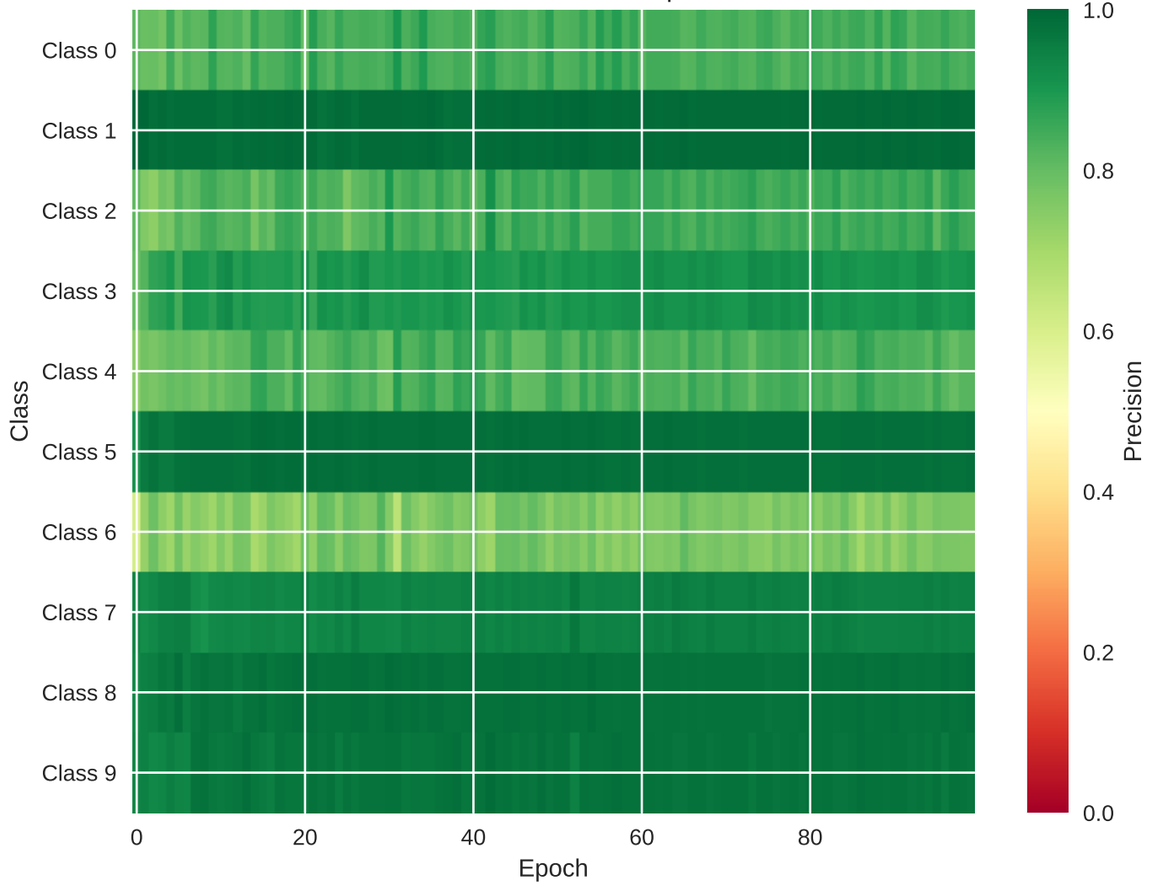


Per-Class Precision Analysis

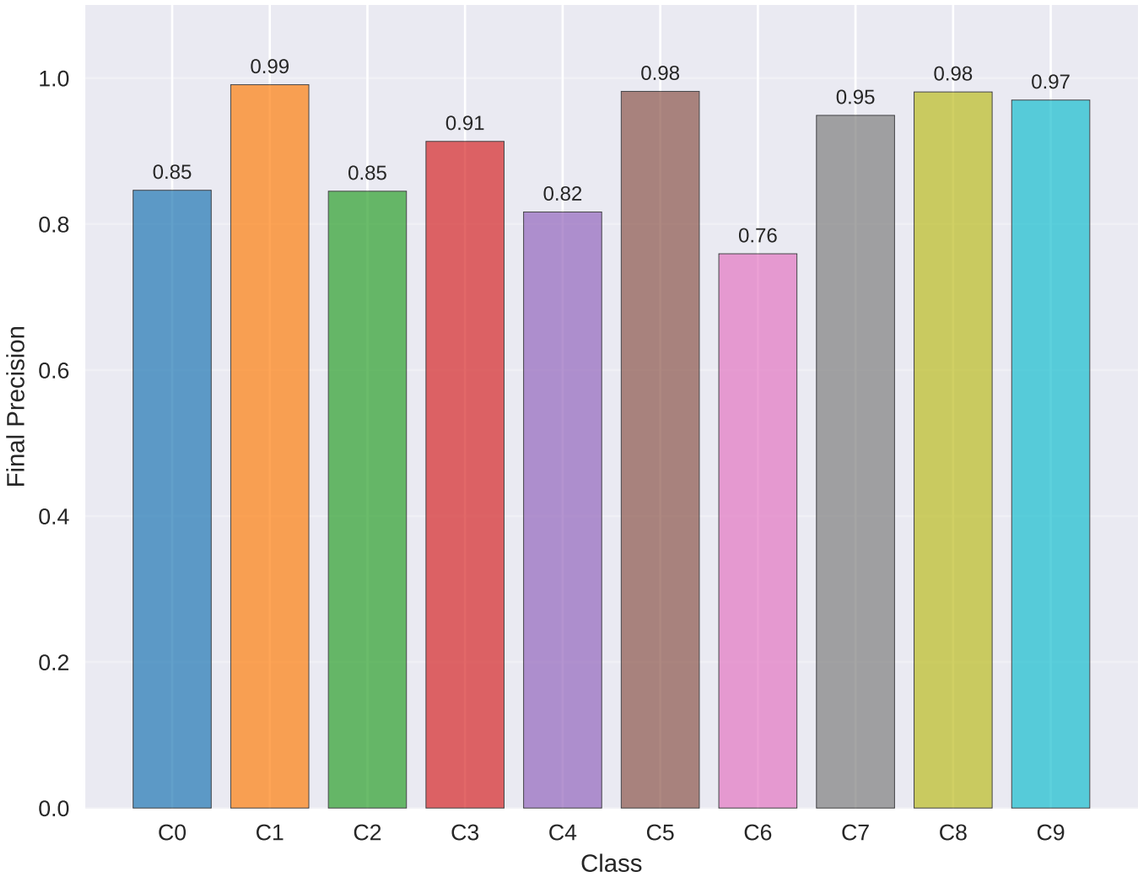
□ Per-Class Precision Evolution



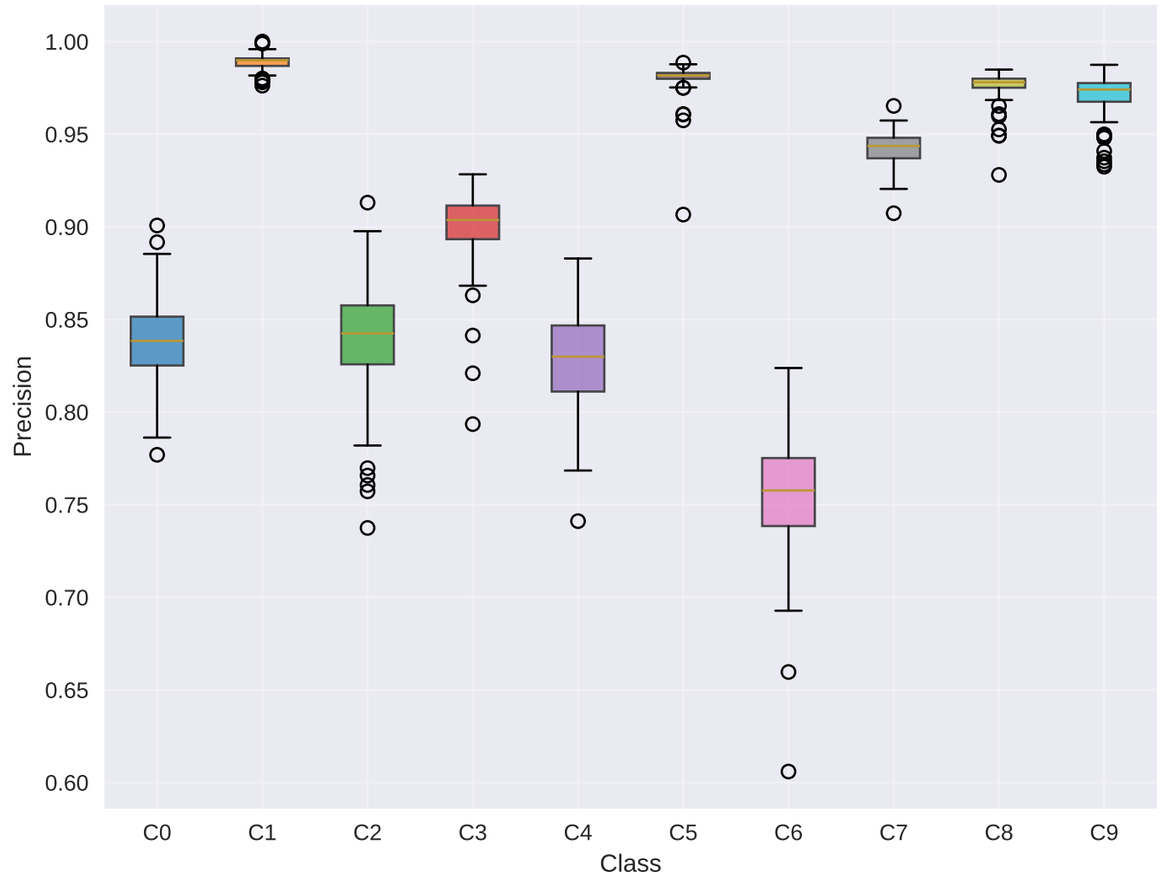
□ Precision Heatmap



□ Final Precision per Class

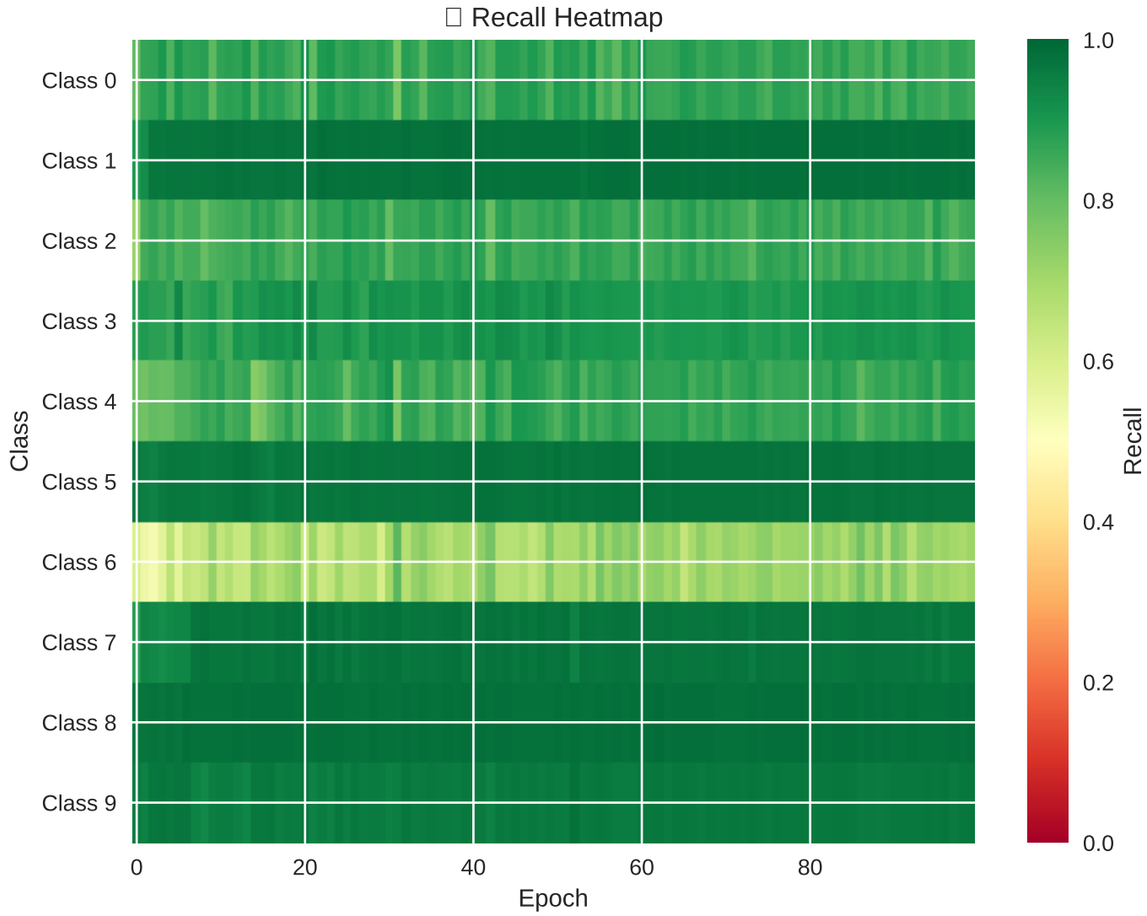


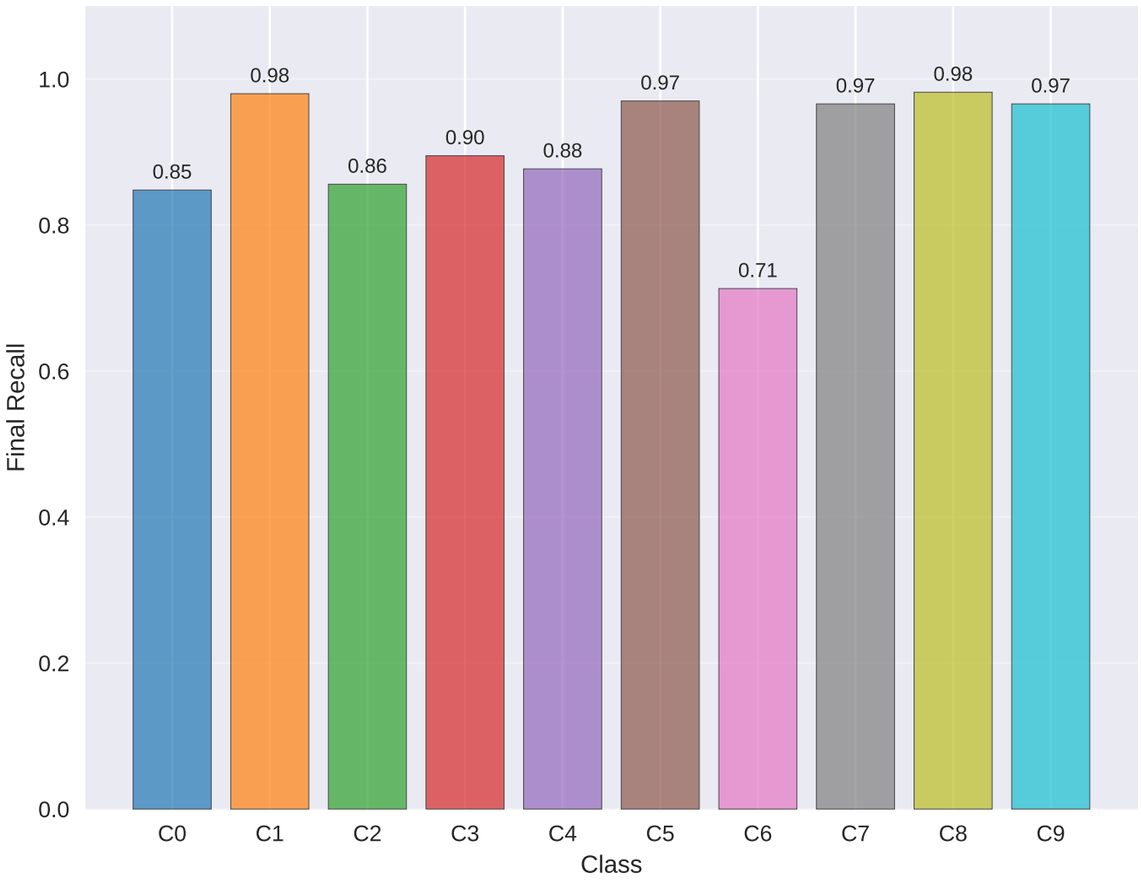
□ Precision Distribution per Class

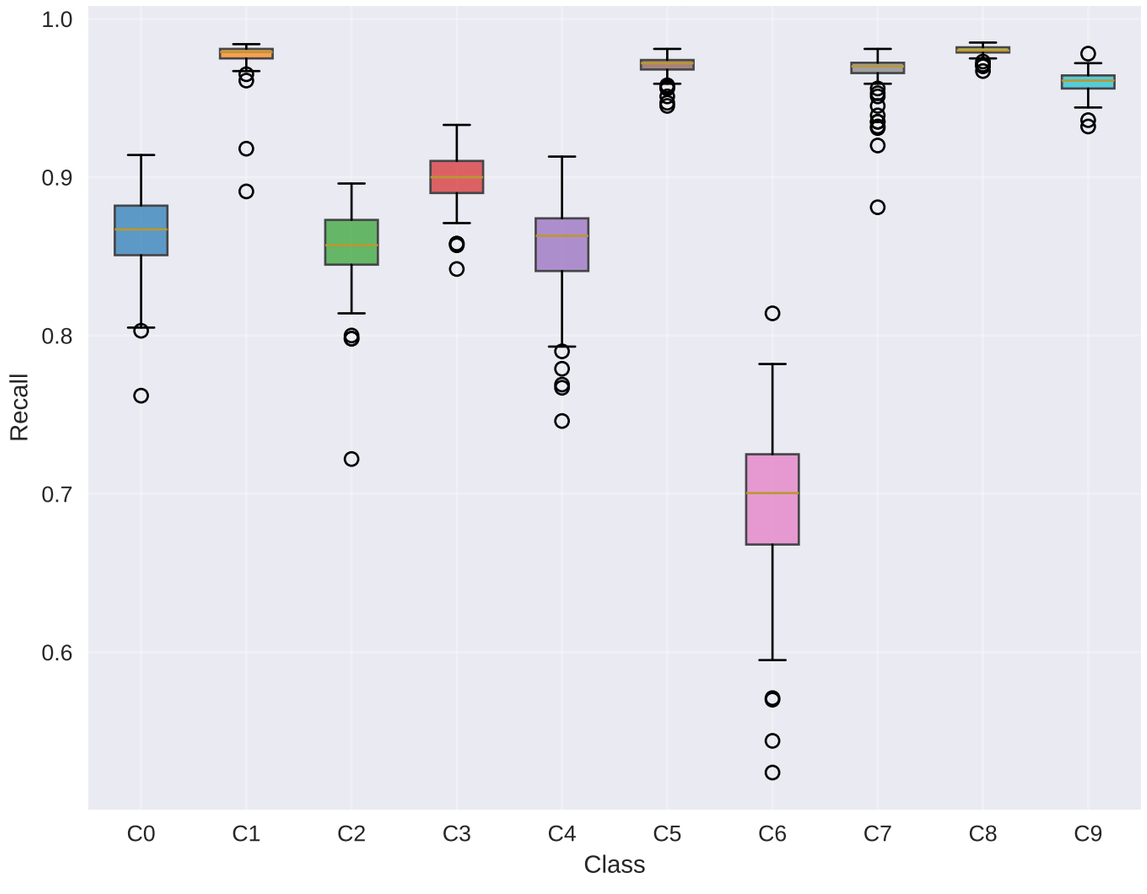


Per-Class Recall Analysis

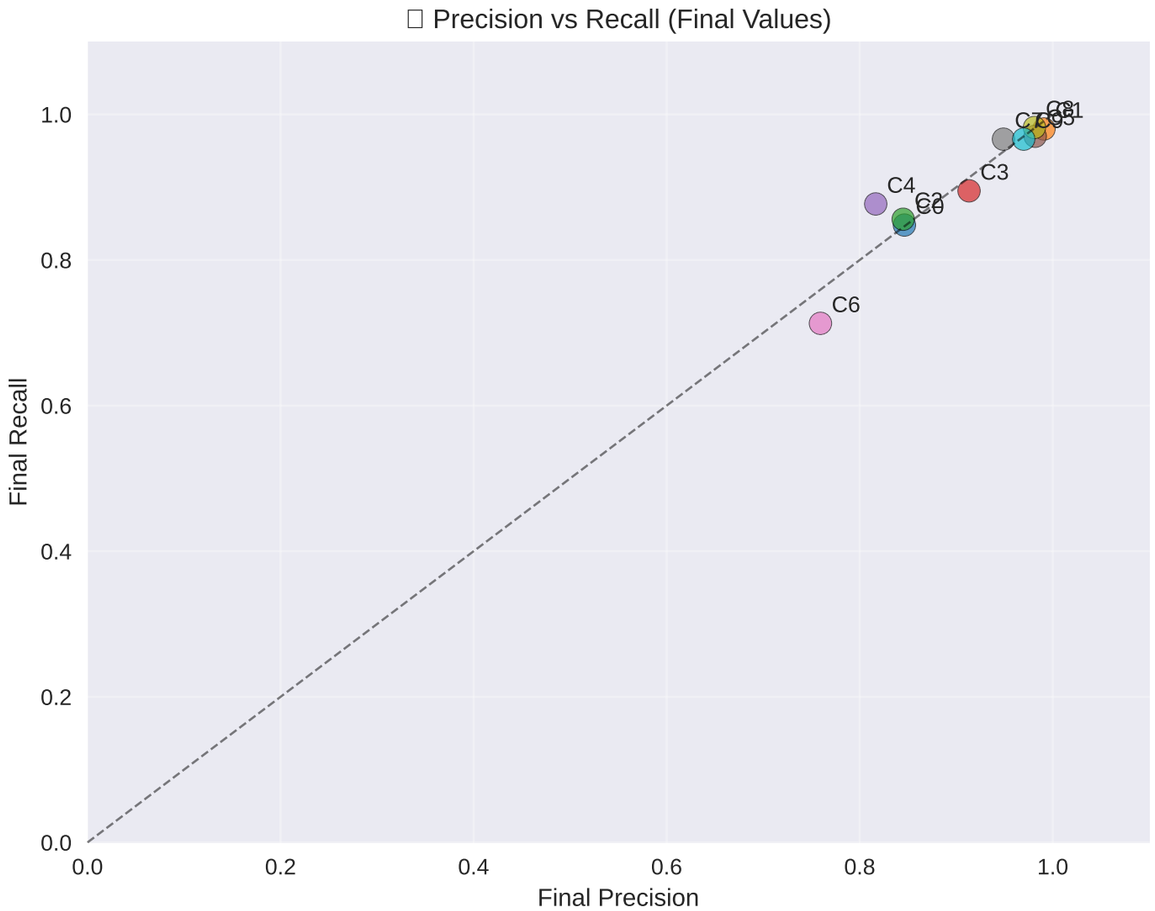




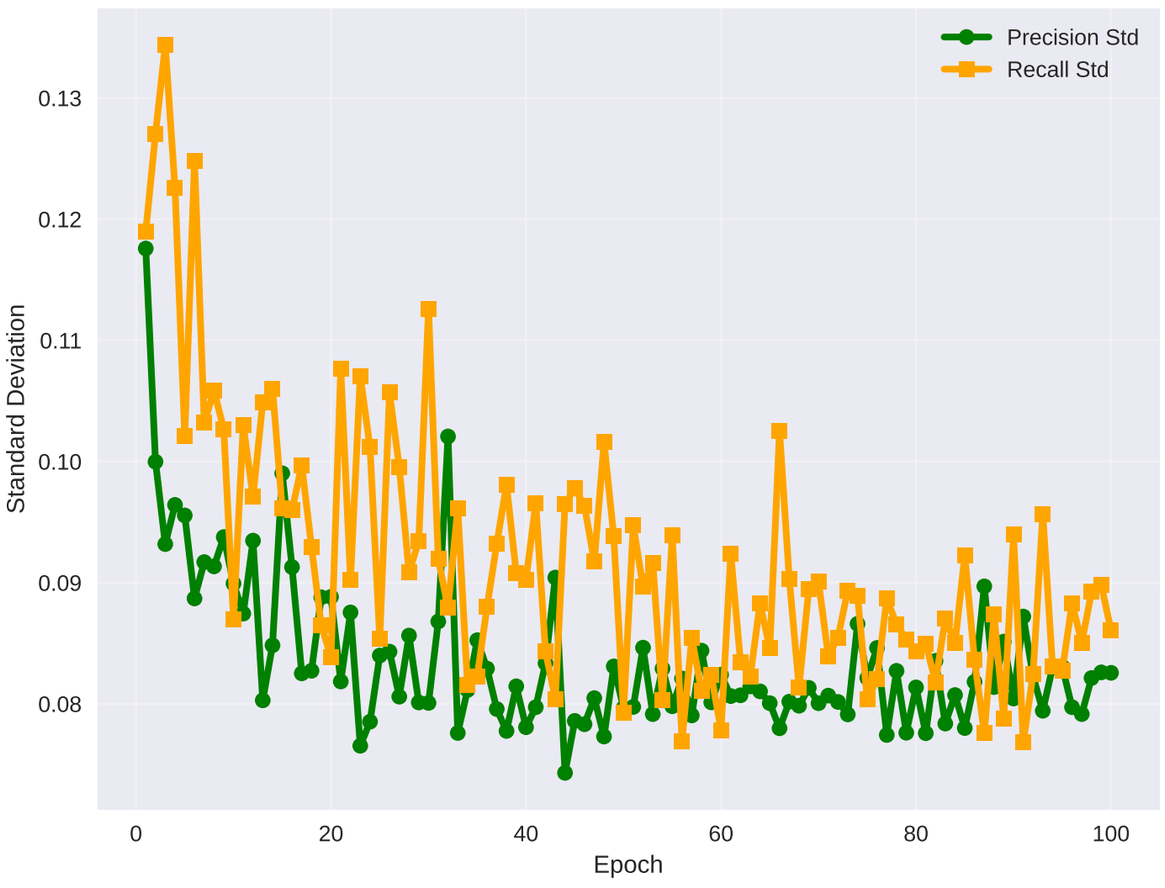


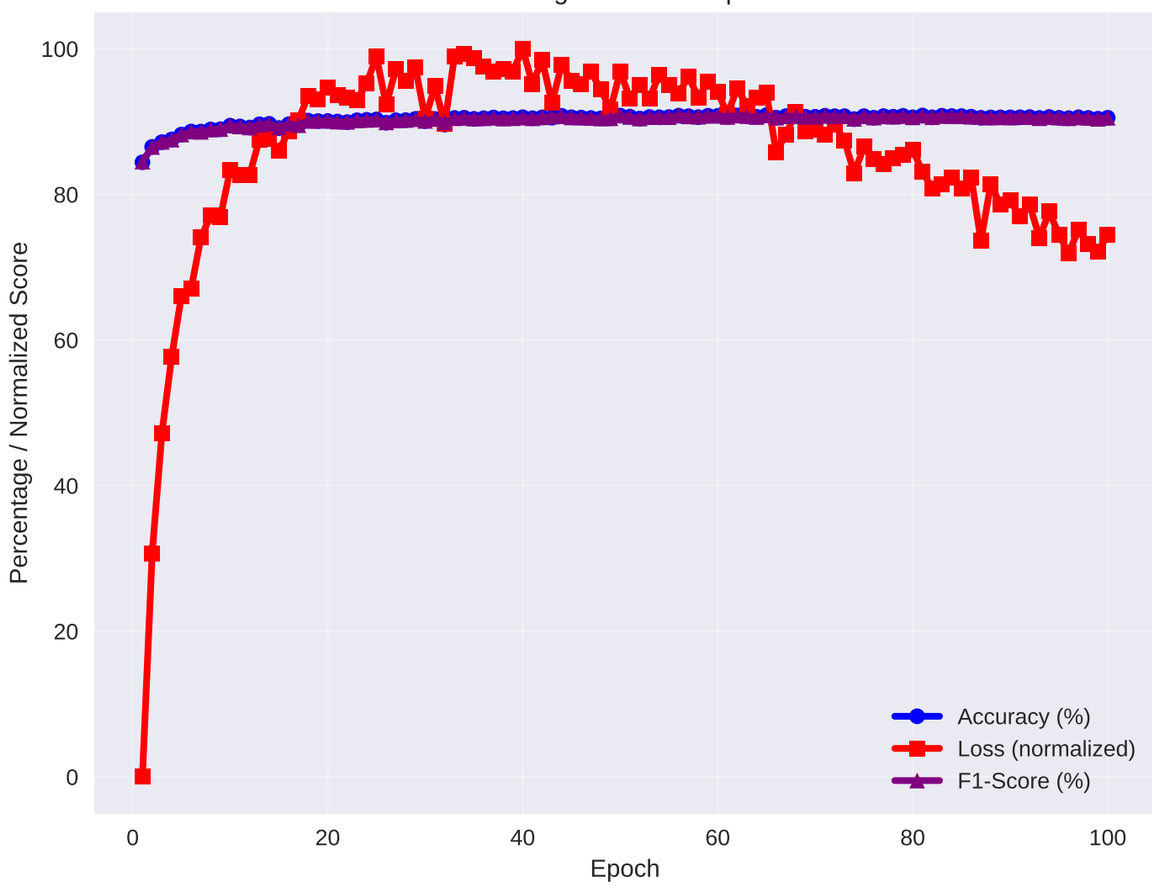


Comparative Analysis









Performance Summary Report

```
FEDERATED LEARNING - PERFORMANCE SUMMARY
========================================

FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-31 23:27:27
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.53%
• Loss: 3.1350
• Average Precision: 0.905
• Average Recall: 0.905
• Average F1-Score: 0.905

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.11%
• Riduzione Loss: +1.3300
• Stabilità Accuracy (CV): 0.0108

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.991)
• Peggiore Precision: Classe 6 (0.759)
• Migliore Recall: Classe 8 (0.982)
• Peggiore Recall: Classe 6 (0.713)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.846, R=0.848, F1=0.847
• Classe 1: P=0.991, R=0.980, F1=0.985
• Classe 2: P=0.845, R=0.856, F1=0.850
• Classe 3: P=0.913, R=0.895, F1=0.904
• Classe 4: P=0.817, R=0.877, F1=0.846
• Classe 5: P=0.982, R=0.970, F1=0.976
• Classe 6: P=0.759, R=0.713, F1=0.735
• Classe 7: P=0.949, R=0.966, F1=0.957
• Classe 8: P=0.981, R=0.982, F1=0.982
• Classe 9: P=0.970, R=0.966, F1=0.968

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
```